

Bank Loan Risk Management Analysis - README

Bank Loan Risk Management Analysis

Overview:

This project analyzes customer loan data to identify factors influencing loan defaults and assess credit risk. Using Python, SQL, Excel, and Power BI, the project generates insights that help financial institutions make informed lending decisions.

Business Problem:

Banks face financial losses when customers fail to repay loans. The objective is to identify high-risk borrowers, analyze repayment patterns, and support better credit risk management.

Tools & Technologies:

- Python (Pandas, NumPy, Matplotlib)
- SQL
- Excel
- Power BI

Project Workflow:

1. Data Cleaning and Validation
2. Exploratory Data Analysis (EDA)
3. SQL-Based Business Analysis
4. Power BI Dashboard Development
5. Business Insights and Recommendations

KPIs:

- Total Customers
- Total Loan Amount
- Average Credit Score
- Default Rate
- Average Debt-to-Income Ratio

Key Insights:

- Lower credit scores are associated with higher default rates.
- High debt-to-income ratios increase repayment risk.
- Late payment history is a strong indicator of default.
- Employment stability influences loan performance.

Skills Demonstrated:

Data Analysis, SQL, Python, Power BI, Data Visualization, Business Intelligence, Credit Risk Analytics, and Problem Solving.